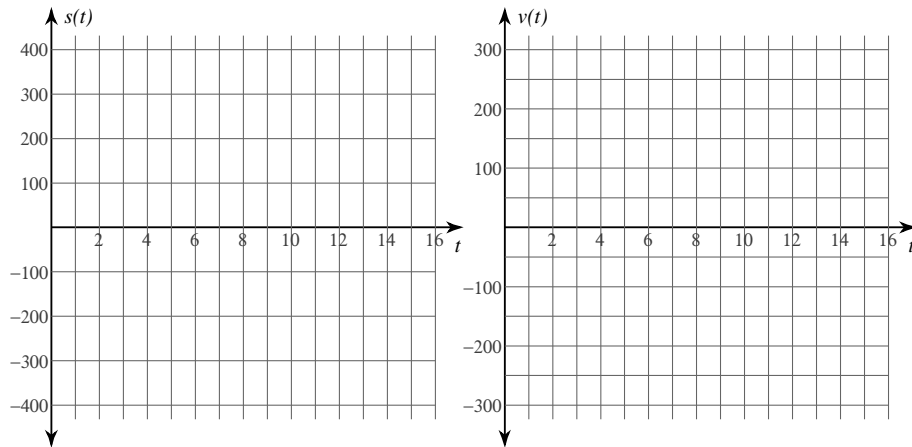


A particle moves along a horizontal line. Its position function is $s(t)$ for $t \geq 0$. For each problem, find the maximum speed and times t when this speed occurs over the given interval. You may use the blank graphs to sketch $s(t)$ and $v(t)$.

4) $s(t) = -t^3 + 18t^2 - 81t$; $2 \leq t \leq 7$



A particle moves along a horizontal line. Its position function is $s(t)$ for $t \geq 0$. For each problem, find the velocity function $v(t)$, the acceleration function $a(t)$, the times t when the particle changes directions, the intervals of time when the particle is moving left and moving right, the times t when the acceleration is 0, and the intervals of time when the particle is slowing down and speeding up. You may use the blank graphs to sketch $s(t)$, $v(t)$, and $a(t)$.

5) $s(t) = t^3 - 23t^2 + 120t$

